

Kimi-Audio: (source: *Kimi Audio Technical Report*)

1. Core Architecture overview:

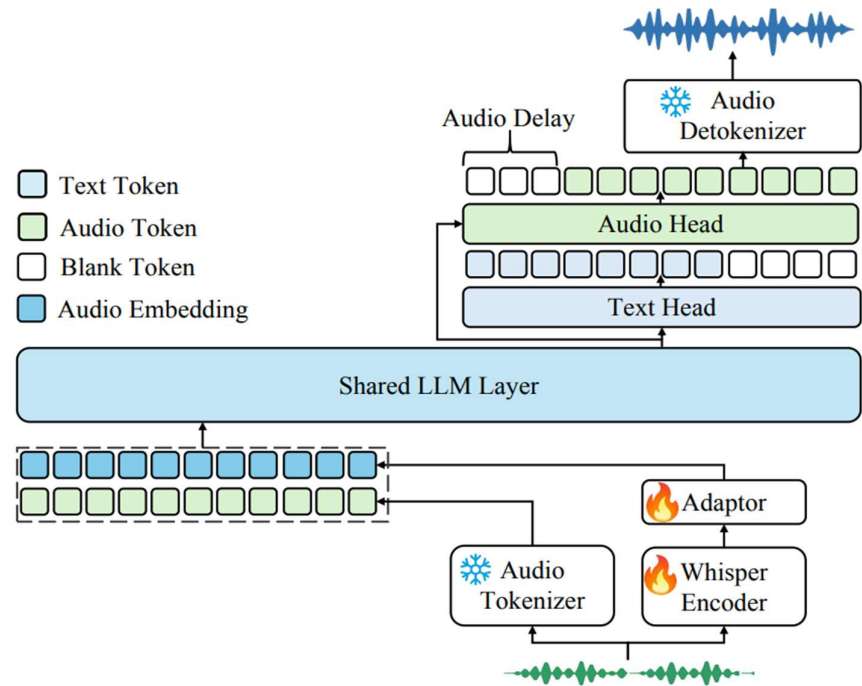


Figure 2: Overview of the Kimi-Audio model architecture: (1) an audio tokenizer that extracts discrete semantic tokens and a Whisper encoder that generates continuous acoustic features; (2) an audio LLM that processes audio inputs and generates text and/or audio outputs; (3) an audio detokenizer converts audio tokens into waveforms.

2. Real-time client-server communication

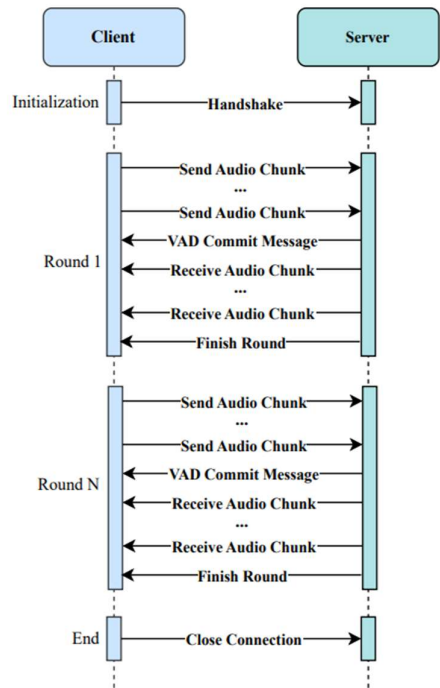


Figure 4: The client-server communication for real-time speech-to-speech conversation in Kimi-Audio.

3. Production Deployment for real-time speech to speech process

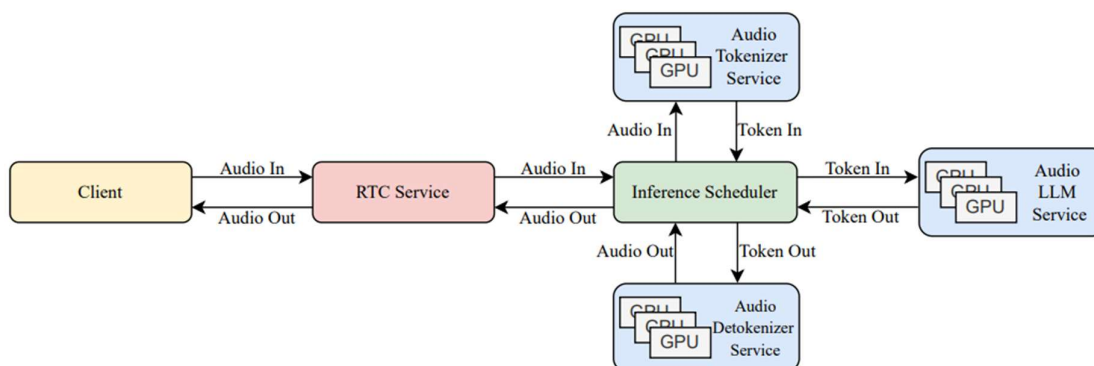
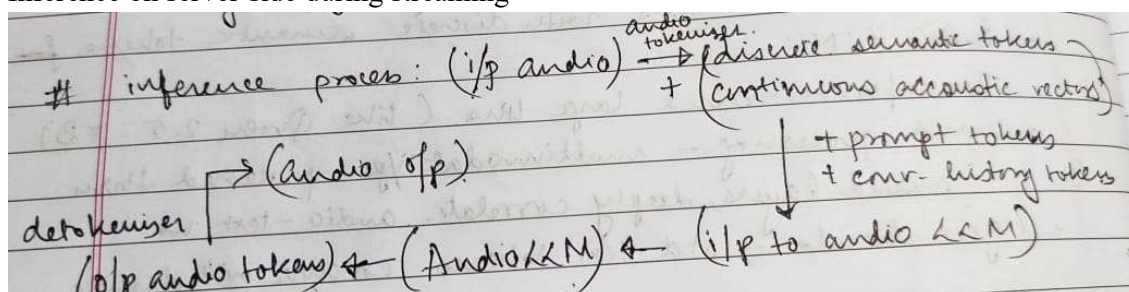


Figure 5: The workflow of production deployment for real-time speech-to-speech conversation in Kimi-Audio.

4. Inference on server-side during streaming



5. Processing Pipeline for audio pre-training data

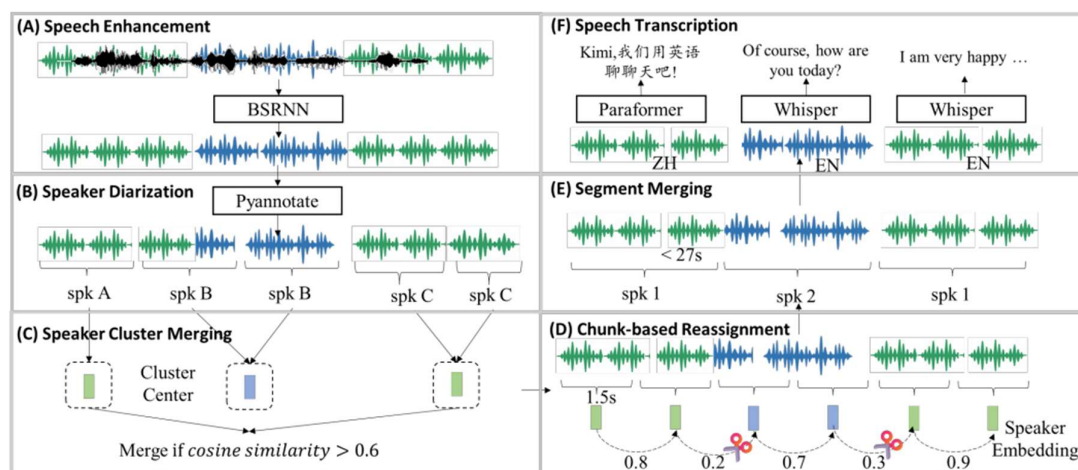


Figure 3: Processing pipeline for the audio pre-training data.

6. Post Training- STFT data pipeline

Fig 3 : post-training / SFT Data pipeline

- base kuni-audio does not inherently know complex cardio vocabs, empathetic clinical bedside manner, regional Indian lang. → adapt & customizing
- Hinglish / Indian lang → shared LLM core → pre-train with text-audio pairs of Indian regional lang
- Cardio, Clinical vocabs - (Swen-2.5) → multilingual token-embedding, SFT head → inject ~~medical~~ ^{medical} knowledge graphs during SFT alignment phase
- Pathological / Noisy Speech → Continuous acoustic adaptor → augment i/p data (whisper) with clinical noise, shurred speech patterns
- Edge device streaming → Flow-matching detokenizers & Vocoders → compress 11 heads using quantization to fit on-device edge AI chips

7. Final proposed design pipeline

